

EXPERIMENT NO. 2(a)

Aim:-Prepare the connection and testing of a fluorescent tube light.

Apparatus Required:-

S. No.	Name	Range	Quantity
1	Fluorescent tube	40W, 230V, 50Hz, 120cm	1
2	Choke coil	40W, 230V, $\cos \phi = 0.5$	1
3	Starter	Glow type for 20/40/65/80W tube	2
4	Test lamp	15W/100W/200W	1/1/1
5	Connecting leads	-----	As per required

Theory:-

A fluorescent lamp is a gas-discharge lamp that uses electricity to excite mercury vapor. The excited mercury atoms produce short-wave ultraviolet light that then causes a phosphor to fluoresce, producing visible light. A fluorescent lamp converts electrical power into useful light more efficiently than an incandescent lamp. Lower energy cost typically offsets the higher initial cost of the lamp. The lamp fixture is more costly because it requires a ballast to regulate the current through the lamp.

A fluorescent lamp contains following parts:-

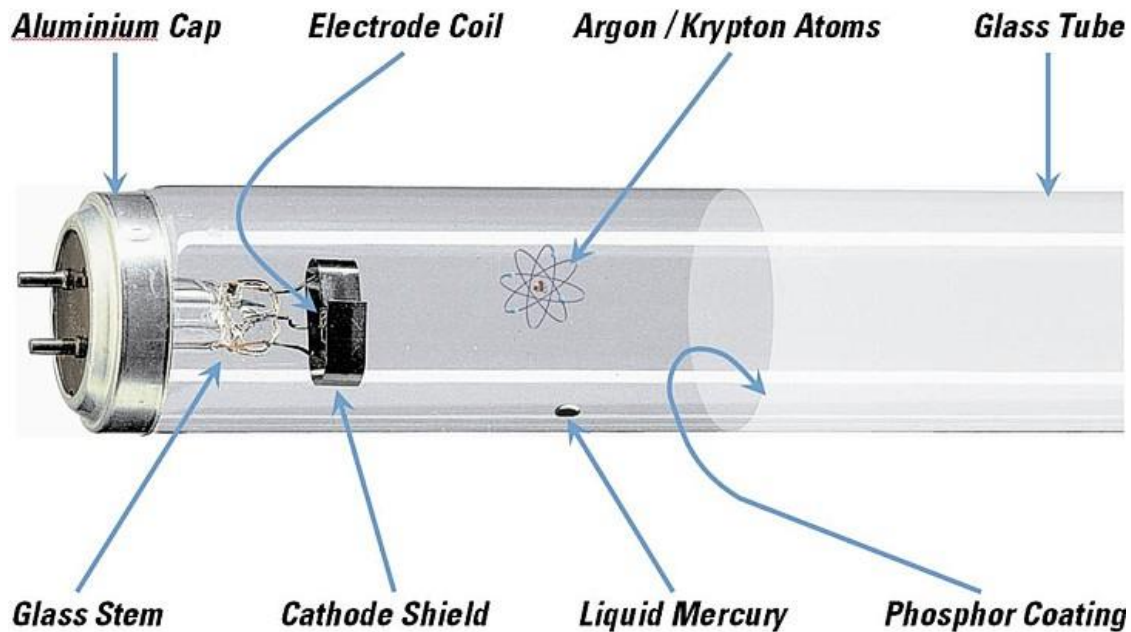
Construction:-

- (1) Fluorescent tube
- (2) Choke or ballast
- (3) Starter

(1) Fluorescent tube:-

A fluorescent lamp tube is filled with a gas containing low pressure mercury vapor and argon, xenon, neon, or krypton. The pressure inside the lamp is around 0.3% of atmospheric pressure. The inner surface of the bulb is coated with a fluorescent (and often slightly phosphorescent) coating made of varying blends of metallic and earth phosphor salts. The bulb's electrodes are typically made of coiled tungsten and usually referred to as cathodes because of their prime function of emitting electrons. For this, they are coated with a mixture

of barium, strontium and calcium oxides chosen to have a low emission temperature. Compact fluorescent lamps have several small-diameter tubes joined in a bundle of two, four, or six, or a small diameter tube coiled into a spiral, to provide a high amount of light output in little volume.



Choke or ballast:-

Instant start ballast starts lamps without heating the cathodes at all by using high voltage (around 600 V). It is the most energy efficient type, but gives the least number of starts from a lamp as emissive oxides are blasted from the cold cathode surfaces each time the lamp is started. This is the best type for installations where lamps are not turned on and off very often. Rapid start ballast applies voltage and heats the cathodes simultaneously. It provides superior lamp life and more cycle life, but uses slightly more energy as the cathodes in each end of the lamp continue to consume heating power as the lamp operates. A dimming circuit can be used with dimming ballast, which maintains the heating current while allowing lamp current to be controlled. Programmed-start ballast is a more advanced version of rapid start. This ballast applies power to the filaments first, then after a short delay to allow the cathodes to preheat, applies voltage to the lamps to strike an arc. This ballast gives the best life and most starts from lamps, and so is preferred for applications with very frequent power cycling such as vision examination rooms and restrooms with a motion detector switch.

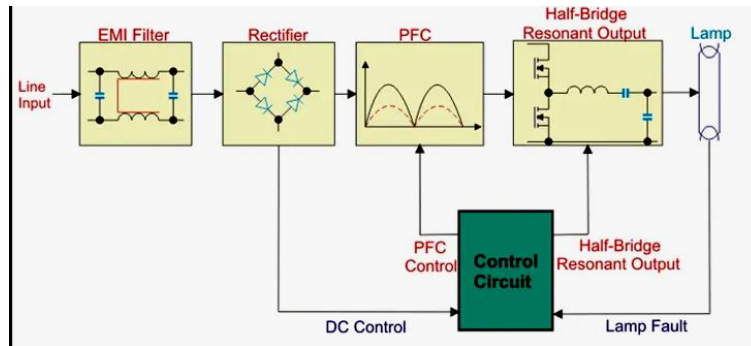


Fig. Electronics ballast Choke Circuit

Working Principle of an Electronic Ballast

Electronic ballast takes supply at 50 – 60 Hz. It first converts AC voltage into DC voltage. After that, filtration of this DC voltage is done by using a capacitor configuration. Now filtered DC voltage is fed to the high-frequency oscillation stage where oscillation is typically square wave and frequency range are from 20 kHz to 80 kHz.

Hence output current is with very high frequency. A small amount of inductance is provided to be associated with a high rate of change of current on high frequency to generate a high value. Generally, more than 400 V is required to strike the gas discharge process in fluorescent tube light. When the switch is ON, the initial voltage across the lamp becomes 1000 V around due to high value, hence gas discharge takes place instantaneously. Once the discharge process is started, the voltage across the lamp is decreased below 230V up to 125V and then this electronic ballast allows limited current to flow through this lamp.

This control of voltage and current is done by the control unit of the electronic ballast. In the running condition of fluorescent lamps, electronic ballast acts as a dimmer to limit current and voltage.

Starter-:

The automatic glow starter shown in the photograph to the left consists of a small gas-discharge lamp, containing neon or argon in parallel with a normally-open bi-metallic switch. It functions as a time-delay switch, turning the filaments of the fluorescent tube on for a few seconds to start it, and then turning them off. When power is first applied to the circuit, a glow discharge will appear over the electrodes in the starter lamp. This glow discharge will heat the gas in the starter and cause the bi-metallic contact to bend towards the other contact. When the contacts touch, the two filaments of the fluorescent lamp and the ballast will effectively be switched in series to the supply voltage. The current through the filaments causes them to heat up and emit electrons into the tube gas by thermionic emission

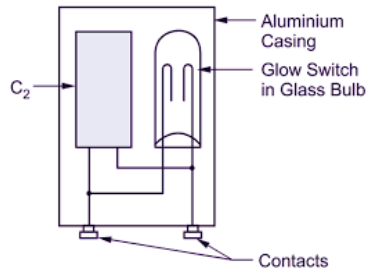


Fig. Starter Circuit

Connection and working of fluorescent lamp:-

Here we connect one ballast, and one switch and the supply is series as shown. Then we connect the fluorescent tube and a starter across it.

- When we switch ON the supply, full voltage comes across the lamp and as well as across the starter through the ballast. But at that instant, no discharge happens, i.e., no lumen output from the lamp.
- At that full voltage first the glow discharge is established in the starter. This is because the electrodes gap in the neon bulb of starter is much lesser than that of the fluorescent lamp.
- Then gas inside the starter gets ionized due to this full voltage and heats the bimetallic strip. That causes to bend the bimetallic strip to connect to the fixed contact. Now, current starts flowing through the starter. Although the ionization potential of the neon is more than that of the argon but still due to small electrode gap, a high voltage gradient appears in the neon bulb and hence glow discharge gets started first in the starter.
- As soon as the current starts flowing through the touched contacts of the neon bulb of the starter, the voltage across the neon bulb gets reduced since the current, causes a voltage drop across the inductor(ballast). At reduced or no voltage across the neon bulb of the starter, there will be no more gas discharge taking place and hence the bimetallic strip gets cool and breaks away from the fixed contact. At the time of breaking of the contacts in the neon bulb of the starter, the current gets interrupted, and hence at that moment, a large voltage surge comes across the inductor(ballast).

$$V = L \frac{di}{dt}$$

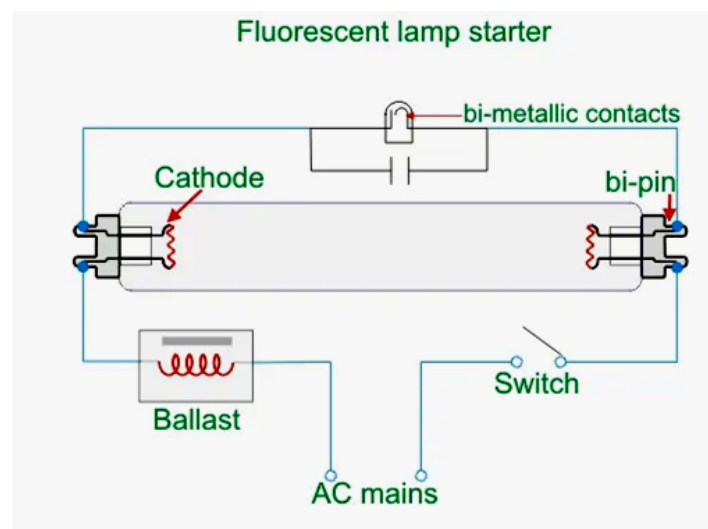
Where, L is inductance of inductor

and $\frac{di}{dt}$ is rate of change of current.

- This high valued surge voltage comes across the fluorescent lamp (tube light) electrodes and strikes penning mixture (mixture argon gas and mercury vapor).

- Gas discharge process gets started and continues and hence current again gets a path to flow through the fluorescent lamp tube (tube light) itself. During discharging of penning gas mixture the resistance offered by the gas is lower than the resistance of starter.
- The discharge of mercury atoms produces ultraviolet radiation which in turn excites the phosphor powder coating to radiate visible light.
- Starter gets inactive during glowing of fluorescent lamp (tube light) because no current passes through the starter in that condition.
- When a sufficiently high voltage is applied across the electrodes, a strong electric field is set up. A small amount of current through the electrodes filaments heats up the filament coil. As the filament is oxide coated, a sufficient amount of electrons is produced, and they rush from the negative electrode or cathode to the positive electrode or anode due to this strong electric field. During the movement of free electrons, the discharge process gets established.

Circuit Diagram:-



ADVANTAGE:-

1. Longer life of about 7500 working hours.
2. High efficiency about 40 lumes/watt.
3. It is economical in use in long run.
4. Available in different colors.

DISADVANTAGE:-

1. Its initial cost is too high.
2. Fixing of its accessories takes time.

3. Its power factor is low.

Procedure:-

1. Make the connection as per as diagram.
2. Connections check and give the supply.
3. When lamp does not glow then filament open.
4. When lamp bright glow then filament is short.
5. When dim glow then filament is in proper working condition.

Results:-we have successfully connection and study of fluorescent lamp and testing of testing of its various parts has been done.

Viva- Voice:-

Que .1) Filament of lamp is made by....?

Que 2) What is the use of inert gas in lamps?

Que3) Why filament is in coil shape?

Que -4) What is the composition of inert gases used in filament lamp?

Que.-5)What is life of approx. incandesend lamp?